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Live Demo - CS5100-Live Demo .mp4
GitHub - https://github.com/pinkcodingmuscle/phising_ga_paper.git

Enhancing Cybersecurity through Genetic Algorithm-Based Phishing Website Detection

This project was completed collaboratively with a clear division of responsibilities. Esther Mukuye handled data preprocessing, dataset preparation, and implementation of the genetic algorithm for feature selection. Marcela Perro focused on model development, evaluation, and performance comparison. Both contributors worked together on result analysis and the final report and presentation.

Abstract

Phishing remains a major cybersecurity threat that exploits human behavior to obtain sensitive information, while traditional detection methods struggle to adapt to evolving attack strategies. This study reproduces and evaluates a hybrid approach proposed by Biplob et al. (2024), which combines Genetic Algorithms (GA) for feature selection with machine learning (ML) classifiers.

We implement a modular pipeline consisting of preprocessing, GA-based feature selection, and classification using multiple models. Our objective is to assess the reproducibility of the paper's reported accuracy of 99.99%.

Our results show strong performance, reaching up to 97.53% accuracy, but fail to replicate the reported results. This discrepancy highlights the impact of stochastic processes and the importance of transparency in experimental design. Overall, GA-based feature selection is effective, but the reported performance is not consistently reproducible.

1. Introduction

Phishing is a social engineering attack that manipulates users into revealing sensitive information through deceptive emails, messages, or websites. Unlike traditional cyberattacks, phishing exploits human behavior, making it one of the most prevalent and costly cybersecurity threats.

Traditional detection methods such as blacklisting rely on known patterns and struggle to detect newly generated attacks. This limitation motivates the use of machine learning approaches that can generalize to unseen threats.

This study reproduces the methodology proposed by Biplob et al. (2024), which combines Genetic Algorithms with machine learning classifiers to improve phishing detection. The original study reports near-perfect accuracy (99.99%), which motivates an investigation into the reproducibility of these results.

Our objective is to replicate the pipeline and evaluate whether similar performance can be achieved under independent implementation.

2. Paper Overview

The research paper "Enhancing Cybersecurity through Genetic Algorithm-Based Phishing Website Detection" addresses the critical need for advanced security measures to counter the sophisticated tactics cybercriminals use to trick users into revealing sensitive data. The core idea of the proposed method is a hybrid framework that utilizes the evolutionary optimization power of Genetic Algorithms (GA) to refine the feature selection process for various Machine Learning (ML) classifiers. By simulating natural selection through iterative cycles of initialization, fitness evaluation using an SVM classifier, selection, crossover, and mutation, the GA identifies the most discriminative website characteristics. This optimized feature set is then utilized by machine learning models to differentiate more effectively between authentic and potentially malicious URLs, overcoming the limitations of static traditional detection methods like blacklists.

The authors claim that a GA-based feature selection significantly enhances detection precision, achieving a near-perfect accuracy rate of 99.99% across multiple datasets. This research advances the practical application of genetic algorithms in phishing detection, showcasing a model that is both adaptable and capable of generalizing well to untested data. Furthermore, by identifying high-performing feature subsets, the study provides a robust methodology for improving the efficacy of existing cybersecurity frameworks against quickly evolving phishing threats.

To validate these claims, the experimental setup involved a rigorous process using three real-world datasets: the Mendeley Phishing Dataset and two variants of the UCI Phishing Website Dataset. The methodology included extensive data preprocessing, such as handling missing values, normalizing numerical features via z-score normalization, and managing outliers to prevent any single feature from unfairly dominating the GA process. The experiments were conducted in a stable computing environment where the GA was configured with specific parameters, and its results were compared against 12 different classification algorithms and several previously published studies. Performance was measured using standard industry metrics, including accuracy, precision, recall, and F1-score, to ensure a comprehensive evaluation of the hybrid model's effectiveness.

3. Theoretical Framework

3.1 Genetic Algorithms

Genetic Algorithms are stochastic optimization techniques inspired by natural selection. Each solution is represented as a binary vector indicating selected features.

The GA operates through:

- **Population:** Set of candidate solutions (chromosomes)
- **Selection:** Tournament selection (k = 3)
- **Crossover:** Single-point crossover (rate = 0.8)
- **Mutation:** Bit-flip mutation (rate = 0.01)

These operations allow the GA to efficiently explore high-dimensional feature spaces.

3.3 Classifier Models

The purpose of using these algorithms is to find the best way to classify malicious versus legitimate websites to see which one performs best on our data.

1. ***Support Vector Machine (SVM)*** - Maximizes margin between classes
2. ***Decision Trees*** - Captures nonlinear decision boundaries
3. ***Logistic Regression*** - Baseline linear model

4.Methodology

This study follows a reproduction-based approach, implementing the methodology described by Biplob et al. (2024) to evaluate the effectiveness of Genetic Algorithm (GA)-based feature selection for phishing website detection.

4.1 Dataset and Features

We use the same datasets as the original study:

- Mendeley Phishing Dataset (10,000 instances, 48 features)
- UCI Phishing Dataset (88,647 instances, 111 features)
- UCI Subset (58,645 instances, 111 features)

These datasets include both phishing and legitimate websites and vary in size and feature dimensionality.

4.2 Data Preprocessing

To ensure consistency with the original methodology, the dataset undergoes the following preprocessing steps:

- Missing values are handled using mean imputation
- Features are normalized using z-score standardization
- The dataset is split into training and testing sets using a 70/30 ratio with stratification

To ensure consistent model performance across multiple datasets, feature scaling is applied using z-score standardization (StandardScaler), which transforms each feature to have zero mean and unit variance. This step is critical because features may have significantly different numerical ranges (e.g., URL length versus binary indicators), and without normalization, models may become biased toward features with larger magnitudes.

Since this study utilizes multiple datasets with varying numbers of features, preprocessing is applied independently to each dataset. Missing values are handled using SimpleImputer (mean strategy), and scaling is performed after imputation to ensure all features contribute comparably to the classification process, regardless of dataset size or structure.

To prevent data leakage, the scaler is fitted on the training data and then applied to the corresponding test set.

4.3 Genetic Algorithm Configuration

Feature selection is performed using a wrapper-based Genetic Algorithm. Each chromosome is represented as a binary vector, where each gene indicates whether a feature is selected.

The GA operates using the following configuration:

- Population size - **50**
- Number of generations - **100**
- Selection method - **Tournament selection (k = 3)**
- Crossover - **Single-point crossover (rate = 0.8)**
- Mutation - **Bit-flip mutation (rate = 0.01)**

In our GA implementation, we use tournament selection (k = 3) as the primary mechanism for selecting parent chromosomes. At each step, three individuals are randomly sampled from the population and evaluated based on their fitness scores. Fitness is defined as the cross-validated performance of an SVM classifier trained on the corresponding feature subset.

The individual with the highest fitness score is selected as a parent for the next generation. This approach promotes high-quality solutions while maintaining diversity within the population.

Each individual is evaluated using a classifier, where fitness is defined as the classification performance (accuracy/F1-score) obtained through cross-validation.

To reduce computational complexity, a mutual information-based prefiltering step is applied prior to GA optimization, limiting the search space to the top k informative features.

4.4 Model Training and Evaluation

Following feature selection, classifiers including Support Vector Machine (SVM), Decision Tree, and Logistic Regression are trained using the selected feature subsets.

Model performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

The results are then compared directly to those reported in the original paper to assess reproducibility.

The fitness of each chromosome is evaluated based on the classification performance of the selected feature subset. Specifically, we use 5-fold cross-validation with an SVM classifier and define fitness as the mean accuracy across folds.

This wrapper-based evaluation ensures that feature subsets are assessed based on their direct impact on model performance.

5. Results

The primary objective of this study is to reproduce the results reported by Biplob et al. (2024) and evaluate how closely their claimed performance can be achieved under an independent implementation.

To ensure a fair comparison, we maintained consistency with the original study by using the same datasets and identical GA configuration parameters, including population size, mutation rate, crossover rate, and selection strategy. Despite this, we were unable to replicate the reported near-perfect accuracy of **99.99%**.

5.1 UCI Dataset (small) [7]

Model performance remained strong but significantly below the reported benchmark. The Decision Tree model achieved the highest accuracy of **92.82%**, followed by SVM at **90.74%**, and Logistic Regression at **88.14%**. Across all models, precision, recall, and F1-scores were closely aligned with accuracy, indicating balanced classification performance without significant bias toward false positives or false negatives.

5.2 UCI Dataset(full) [7]

Performance improved on the larger dataset with the Decision Tree model performing best with an accuracy of **94.91%**, followed by SVM at **94.18%**, and Logistic Regression at **91.92%**. The consistency across all evaluation metrics indicates stable model behavior and effective feature selection.

5.3 Phishing Dataset [5]

The highest performance was observed on Phishing Dataset [5], where the SVM achieved an accuracy of **97.53%**, closely followed by Decision Tree at **96.57%**, and Logistic Regression at **94.33%**. These results represent the closest approximation to the performance reported by the authors, though still below the claimed 99.99% accuracy.

A summary of model performance across all datasets is shown in Table 2.

Table 2. Model Performance Across Datasets

Dataset	***Model***	***Accuracy***	***Precision***	***Recall***	***F1 score***
UCI Small	Decision Tree	92.82%	92.84%	92.82%	92.81%
UCI Small	SVM	90.74%	90.75%	90.74%	90.73%
UCI Small	Logistic Regression	88.14%	88.22%	88.14%	88.12%
UCI Full	Decision Tree	94.91%	94.91%	94.91%	94.91%
UCI Full	SVM	94.18%	94.18%	94.18%	94.18%

UCI Full	Logistic Regression	91.92%	92.06%	91.92%	91.97%
Mendeley	Decision Tree	97.53%	97.54%	97.53%	97.53%
Mendeley	SVM	96.57%	96.58%	96.57%	96.57%
Mendeley	Logistic Regression	94.33%	94.35%	94.33%	94.33%

To account for the inherent randomness of Genetic Algorithms, we performed five independent runs using different random seeds. The resulting accuracies showed minimal variance (below **0.05%**), indicating that the GA consistently converges to similar high-quality feature subsets.

This suggests that while exact reproduction of the reported 99.99% accuracy is not achievable, the overall optimization process is stable and reliable.

6. Reproducibility Analysis

Although our results consistently demonstrate high classification performance (**97.53% accuracy)** we were unable to replicate the near-perfect results reported in the original study. This discrepancy persists despite using the same datasets and matching GA hyperparameters.

One likely explanation is the stochastic nature of Genetic Algorithms, where outcomes are highly sensitive to initialization conditions such as the random seed. The original paper does not specify the random seed or provide sufficient implementation details, making exact reproduction infeasible. Additionally, when performance approaches near-perfect accuracy, even minor differences in preprocessing, data splitting, or model implementation can lead to measurable deviations in results.

Overall, our findings suggest that while the proposed GA-based feature selection approach is effective and yields strong performance, the reported results may not be consistently reproducible without full transparency in the experimental setup.

7. Analysis and Discussion

The results of this study highlight both the strengths and limitations of Genetic Algorithm (GA)-based feature selection for phishing detection. Across all datasets, the GA consistently reduced the feature space while maintaining strong classification performance, demonstrating its effectiveness as an optimization technique in high-dimensional settings.

One of the most notable observations is that Decision Trees and SVMs consistently outperformed Logistic Regression. This suggests that nonlinear decision boundaries play a significant role in distinguishing phishing from legitimate websites. Since phishing features often involve complex interactions (e.g., URL structure combined with domain signals), models capable of capturing these relationships naturally perform better.

Despite the strong results, the GA did not produce a dramatic improvement over baseline expectations. While it successfully identified informative feature subsets, the performance gains were incremental

rather than transformative. This suggests that many of the original features are already highly predictive, and the GA primarily improves efficiency rather than drastically boosting accuracy.

A key finding is the gap between our results and the 99.99% accuracy reported in the original paper. Even under matched experimental conditions, such near-perfect performance was not observed. This raises important questions about overfitting, dataset leakage, or undocumented preprocessing steps in the original study. When performance approaches theoretical limits, even minor experimental differences can significantly impact results.

Finally, the stochastic nature of Genetic Algorithms was carefully examined through multiple runs with different random seeds. The minimal variance observed across runs confirms that the GA is stable and reliably converges, suggesting that the discrepancy in results might not be due to randomness but rather differences in implementation details or experimental setup.

Overall, this analysis shows that while GA-based feature selection is a valuable tool, its primary benefit lies in efficient feature reduction and model stability, rather than achieving near-perfect classification performance.

8. Challenges in Reproduction

8.1 Computational Cost

Each individual in a population of 50 must be evaluated in every generation. When combined with 5-fold cross-validation to ensure stability, this results in 250 model trainings per generation. In our experiments on the UCI dataset, a single generation required over 14 hours to complete. Scaling this to the full 100 generations implies a substantial computational burden, which is not explicitly addressed in the original study. This raises concerns about the feasibility and reproducibility of the reported results without access to high-performance computing resources.

8.2 Hyperparameter Ambiguity

Although the original paper specifies key GA parameters such as crossover and mutation rates, it provides limited detail regarding the configuration of the classifiers used during fitness evaluation. Since the GA's performance is directly tied to the underlying model, even minor differences in classifier settings can lead to significant variations in results. To address this, we implemented a modular tuning approach using GridSearchCV for the final models. However, incorporating hyperparameter tuning within the GA pipeline introduces additional computational overhead and further complicates reproducibility.

9. Limitations

Despite the high accuracy achieved by the hybrid GA-ML framework, several limitations must be acknowledged:

9.1 Computational Constraints

Due to the computational demands of SVM training and GA iterations, the number of independent runs was limited. While five runs were sufficient to confirm stability, broader experimentation would require more powerful computational resources.

9.2 Dataset Specificity

Our evaluation is limited to the Mendeley and UCI phishing datasets, which, although widely used, represent a static snapshot of phishing behavior. In practice, phishing techniques evolve rapidly, and feature relevance may shift over time. This limits the generalizability of our findings and suggests that the identified “optimal” feature subsets may not remain effective in future or real-world deployments.

9.3 Time–Optimization Trade-off

To reduce runtime, we applied a mutual information–based prefiltering step before GA optimization. While effective, this may exclude complex feature interactions that a full GA search could capture, potentially limiting optimality.

10. Conclusion and Future Work

This study successfully reproduces the GA-based phishing detection pipeline and demonstrates strong performance, achieving up to 97.53% accuracy. The approach consistently produces stable results across datasets, confirming its effectiveness for feature selection and optimization.

However, the reported accuracy of 99.99% was not reproducible. This highlights the importance of transparency in machine learning research, particularly for stochastic methods where small implementation differences can significantly affect outcomes.

From a practical perspective, GA-based feature selection improves efficiency and helps identify informative features but does not fundamentally overcome limitations of the data or guarantee near-perfect classification.

Future work could include leveraging high-performance computing to expand the search space, incorporating newer datasets to improve generalizability, and exploring hybrid approaches that combine GA with advanced models such as deep learning.

11. Reproducibility Verdict

The methodology proposed by Biplob et al. (2024) **is** partially reproducible**.** While the overall pipeline was successfully replicated and produced consistently strong results, the reported performance was not achieved. The best accuracy observed was 97.53%, indicating a gap likely caused by missing implementation details such as random seeds, preprocessing steps, or model configurations.

Repeated experiments showed minimal variance, confirming that the GA is stable and that the discrepancy is not due to randomness.

Overall, the framework is reproducible, but the reported results are not consistently attainable, emphasizing the need for greater transparency in experimental design.

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